HEALTH INDICA

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Health indicators tell the health story of a nation. Master them!

Health Indicators | PSM | 3rd Year | MBBS

WHAT'S INSIDE

- Classification of Health Indicators p.2
- IMR, NMR, PMR — Components & Significance p.3
- MMR, Life Expectancy & Composite Indices p.4
- DALY, QALY, HDI & Indian Data p.5
- Exam Strategy & High Yield p.6

Classification of Health Indicators

This classification is how every LAQ on this topic should begin!

HEALTH INDICATOR

A variable that can be measured directly to reflect the state of health of a community or population. — Park

1. Mortality Indicators

Indicator	What It Measures	Significance
IMR	Infant deaths (<1 yr) / 1000 LB	BEST overall health indicator
NMR	Neonatal deaths (<28 d) / 1000 LB	Quality of neonatal care
PMR	Perinatal deaths / 1000 total births	Obstetric + neonatal care quality
MMR	Maternal deaths / 100,000 LB	Maternal healthcare quality
CDR	All deaths / 1000 mid-yr pop	Crude — affected by age structure
U5MR	Deaths <5 yr / 1000 LB	Used by UNICEF; child survival
Case Fatality Rate	Deaths from disease / Total cases	Severity of disease

2. Morbidity Indicators

Indicator	Details
Incidence rate	New cases in a time period — measures RISK
Prevalence rate	All existing cases at a point — measures BURDEN
Notification rate	Reported cases of notifiable diseases
Attendance at PHC/OPD	Health service utilisation — indirect morbidity measure
Hospital admission rate	Severity of disease in community

3. Other Categories of Health Indicators

Category	Examples
Nutritional	IMR, proportion of LBW, PEM prevalence, anaemia prevalence
Health care delivery	Doctor:population ratio, bed:population, immunisation coverage
Socioeconomic	Literacy rate, per capita income, HDI, poverty line
Environmental	Safe water access, sanitation coverage, air quality index
Health policy	Health expenditure as % of GDP, out-of-pocket expenditure

IMR is THE most asked indicator in PSM! Know every component cold!

Infant Mortality Rate (IMR) — The Gold Standard

Formula: Deaths of infants < 1 year / Total live births x 1000

India IMR (SRS 2020): 28 per 1000 live births

Why BEST indicator: Sensitive to socioeconomic changes, maternal health, nutrition, environmental conditions, healthcare access. Reflects OVERALL health status of nation.

Components of IMR — Detailed Breakdown

Component	Period	Main Causes	India (SRS 2020)
Early neonatal	0-7 days	Asphyxia, prematurity, LBW	~60% of all IMR
Late neonatal	7-28 days	Sepsis, pneumonia	~15% of all IMR
Post-neonatal	28d - 1 yr	Diarrhoea, ARI, malnutrition	~25% of all IMR

Neonatal Mortality Rate (NMR)

Formula: Deaths < 28 days / Live births x 1000 | India: 20/1000 (SRS 2020)

NMR contributes ~70% of IMR — reducing NMR is KEY to reducing IMR!

Causes: Prematurity #1, birth asphyxia #2, neonatal sepsis #3 (WHO global)

Perinatal Mortality Rate (PMR)

Formula: (Late fetal deaths after 28 wks + Early neonatal deaths) / Total births x 1000

India PMR: ~22/1000 (SRS 2020) — reflects BOTH obstetric and neonatal care quality

Stillbirth = fetal death after 28 weeks gestation (late fetal death)

★ IMR = NMR + Post-NMR | NMR = Early NMR + Late NMR

Neonatal deaths contribute ~70% of IMR. Focus on first 28 days = biggest impact!

Early neonatal (0-7d) is the highest contributor = most critical period.

MMR denominator is per LAKH — not per 1000! Classic trap Q!

Maternal Mortality Rate/Ratio (MMR)

Definition:	Death of a woman while pregnant or within 42 days of termination of pregnancy, from any cause related to pregnancy (NOT accidental)
Formula:	Maternal deaths / Live births x 1,00,000 (per LAKH!)
India MMR:	97/1,00,000 (SRS 2018-20) — declining steadily
SDG target:	< 70/1,00,000 by 2030
Best state:	Kerala (~19) | Worst: Assam (~195)

Causes of Maternal Death in India

Rank	Cause	Details
#1	Hemorrhage (PPH)	Post-partum hemorrhage — most common overall
#2	Hypertensive disorders	Pre-eclampsia, eclampsia — 2nd most common
#3	Sepsis	Puerperal sepsis — infection after delivery
#4	Obstructed labour	Prolonged/neglected labour — rupture uterus
#5	Unsafe abortion	Illegal/unsafe procedures — septic abortion

Life Expectancy at Birth

Definition:	Average number of years a newborn is expected to live
India (SRS):	Male: 69.4 yrs | Female: 72.0 yrs | Combined: ~70.7 yrs
Significance:	Reflects overall health, nutrition, socioeconomic development
Limitation:	Does not reflect quality of life — only quantity (years lived)

Specific vs Crude vs Standardized Rates

Type	Details
Crude rate	Calculated for entire population — simple but misleading (age structure)
Specific rate	Calculated for specific group (age, sex) — more accurate comparison
Standardized rate	Adjusted for age/sex — allows fair comparison between populations

DALY vs QALY = classic comparison MCQ! HDI components are a must!

DALY — Disability Adjusted Life Year

DALY = YLL + YLD

YLL = Years of Life Lost (due to premature death)

YLD = Years Lived with Disability (morbidity burden)

1 DALY = 1 year of healthy life LOST. Used by WHO for Global Burden of Disease.

QALY — Quality Adjusted Life Year

QALY = Years of life x Quality weight (0 to 1)

1 QALY = 1 year of PERFECT health GAINED. Used for cost-effectiveness analysis.

Quality weight: 1 = perfect health, 0 = death. Used in health economics & drug evaluation.

DALY vs QALY — Quick Comparison

Feature	DALY	QALY
Measures	Disease BURDEN (loss)	Health GAIN (benefit)
Direction	Higher = WORSE	Higher = BETTER
Used by	WHO, World Bank	NICE (UK), health economists
Formula	YLL + YLD	Years x Quality weight
Application	Global burden of disease	Cost-effectiveness of treatments

HDI — Human Development Index (UNDP)

★ **HDI = 3 COMPONENTS (Exam Must-Know!)**

1. **Life expectancy at birth**
2. **Mean years of schooling + Expected years of schooling**
3. **GNI per capita (PPP $)**

India HDI 0.633 (2022) — Medium human development (Rank 134/193)

Other Composite Health Indices

Index	Components	Used By
PQLI	Literacy, IMR, LE at age 1	Older index, replaced by HDI
GDI	HDI adjusted for gender gap	UNDP — gender equality
GII	RH, empowerment, labour mkt	Gender Inequality Index — UNDP
HALE	LE adjusted for disability	WHO — healthy life expectancy
GHI	Undernourishment, wasting, stunting	Global Hunger Index

Health indicators are
pure scoring territory!
You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

- 1. Classify health indicators. Describe IMR and its components in detail
- 2. Define MMR. Discuss causes of maternal mortality and prevention strategies
- 3. What is DALY? Differentiate DALY from QALY with applications
- 4. Describe the components of HDI. What is India's current HDI?
- 5. Write about mortality indicators — IMR, NMR, PMR, U5MR with Indian data

Most Repeated SAQs (5-6 marks)

- 1. IMR components and why it is the best health indicator
- 2. DALY formula and interpretation
- 3. HDI — components and India's rank
- 4. MMR calculation and Indian data
- 5. Crude vs specific vs standardized rates
- 6. Life expectancy — definition and Indian values

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Best single indicator of health	IMR
IMR in India (SRS 2020)	28 per 1000 live births
MMR denominator	Per 1,00,000 live births (per lakh)
#1 cause of maternal death in India	Hemorrhage (PPH)
NMR contributes what % of IMR?	~70%
DALY = ?	YLL + YLD (1 DALY = 1 healthy year lost)
QALY: higher value means?	BETTER health (gain)
HDI components	Health (LE) + Education + Income (GNI)
India HDI rank	134/193 (Medium human development)
PQLI replaced by	HDI
U5MR used by which agency?	UNICEF
HALE measures	Healthy life expectancy (LE - disability)

♥ **Behind every indicator is a life that matters, Sammy!**
You're learning to measure health so you can improve it!